

## **PROJECT-I B.E(IT)**



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# **AI-Powered Network Security System with Real-Time ETL Pipelines, Automated MLOps Retraining, and Blockchain-Secured Logs**

## **1. Introduction of the Problem**

**1.1 With the exponential rise in cyberattacks, traditional network defense mechanisms (firewalls, IDS/IPS, antivirus) often fail to detect modern and unknown threats.**

**1.2 Today's networks generate massive amounts of logs, traffic traces, and alerts, but manual monitoring of this "big security data" is slow and inefficient.**

**1.3 Current challenges include:**

- **Difficulty in processing large-scale heterogeneous data in real time.**
- **Low adaptability against zero-day and evolving attack patterns.**
- **Vulnerability of log systems to tampering, which weakens forensic audits.**
- **Absence of an automated compliance-friendly framework for data protection.**

**1.4 This project aims to solve these issues by designing an integrated AI-driven, automated, and blockchain-secured network security system with real-time detection and immutable audit trails.**

## **2. Objectives / Components / Modules of the Project**

### **2.1 Automated ETL Pipeline for Security Data**

- **Extract network logs, firewall data, and traffic details.**
- **Transform and normalize data for ML processing.**
- **Load structured data into secured storage with blockchain-based logging of every ETL operation.**

### **2.2 AI-Powered Threat Detection Module**

- **Train deep learning models (MLP-GRU architectures) for anomaly detection and classification.**

- Detect intrusion attempts, malware, phishing, and other irregular activities with high accuracy.

### **2.3 Blockchain-Secured Audit System**

- Record every ETL operation and system activity on blockchain to create tamper-proof logs.
- Enable forensic auditing and automated regulatory compliance verification.

### **2.4 MLOps for Automation and Scalability**

- Automate retraining and redeployment of ML models as new threats emerge.
- Use model versioning, performance tracking, and drift detection for continuous improvement.

### **2.5 Real-Time Monitoring & Alerts Dashboard**

- Provide a centralized dashboard (using FastAPI/Streamlit) for live threat analytics.
- Generate alerts, prioritize threats, and assist analysts in decision-making.

## **3. Technology Stack**

**3.1 ETL & Processing: Python, Pandas, Apache Airflow, Apache Spark**

**3.2 Data Streaming: Apache Kafka**

**3.3 Machine Learning/AI: TensorFlow, PyTorch, Scikit-learn**

**3.4 MLOps: MLflow, DagsHub**

**3.5 Blockchain: Hyperledger Fabric, Ethereum Smart Contracts, IPFS**

**3.6 Databases & Storage: MongoDB, PostgreSQL, AWS S3**

**3.7 Deployment: Docker, Kubernetes, AWS EC2**

**3.8 Frontend & Monitoring: Streamlit, FastAPI, Grafana**

## **4. Data Used / Data Sources**

### **4.1 Public Security Datasets:**

- CIC-IDS2017 (intrusion detection dataset)
- UNSW-NB15 (network intrusion dataset)
- Phishing Websites Dataset

### **4.2 Real-Time Data Sources:**

- **Logs generated from firewalls, OS event logs, and packet sniffers (Wireshark/TCPdump).**
- **Threat intelligence feeds for emerging attack patterns.**

#### **4.3 Synthetic & Simulated Data:**

- **Attack simulation scenarios to train and validate system.**

## **5. Proposed Methodology (Steps with Flow)**

### **Step 1: Data Ingestion**

**Collect raw data (logs, network packets, firewall alerts).**

### **Step 2: ETL Preprocessing with Blockchain Logging**

**Clean, normalize, and record ETL events on blockchain for auditability.**

### **Step 3: Feature Engineering & Model Training**

**Extract features and train hybrid MLP-GRU deep learning models.**

### **Step 4: Deployment & Inference**

**Deploy trained models in production via MLOps pipeline for real-time predictions.**

### **Step 5: Blockchain-Secured Audit Layer**

**All system events logged in immutable blockchain for compliance.**

### **Step 6: Continuous Monitoring & Alerts**

**System generates real-time alerts via dashboard; ML models retrain automatically using new data.**

### **Step 7: Feedback & Adaptability**

**Analyst feedback + new intelligence updates improve retrained models.**

## 6. Work Plan (Gantt Chart Format)

| Week | Activity                                      | Outcome                                      |
|------|-----------------------------------------------|----------------------------------------------|
| 1-2  | Literature Review & Requirements Analysis     | Problem understanding and scope finalization |
| 3-4  | Data Collection & ETL Pipeline Implementation | Working ETL pipeline with blockchain logging |
| 5-6  | Model Development (MLP-GRU) & Training        | Trained model with baseline accuracy         |
| 7    | MLOps Pipeline Setup                          | Automated training and deployment workflows  |
| 8-9  | Blockchain Integration for Secure Logging     | Immutable and auditable logs ready           |
| 10   | Dashboard Development (FastAPI/Streamlit)     | Real-time monitoring interface               |
| 11   | System Integration & Testing                  | Integrated end-to-end security system        |
| 12   | Optimization and Validation                   | Performance tuning                           |
| 13   | Documentation & Final Report Preparation      | Complete project documentation               |

## 7. Bibliography / References

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